Category Theory for Mathematical Physics — Exercise Sheet 10

Modular tensor categories and anyon models

Variant: Solutions

Conventions as in Lecture 10: $\tilde{S}_{ab} = \sum_c N_{a\bar{b}}^c \, d_c \, \theta_c/(\theta_a\theta_b)$ (Proposition 10.3), $S = \tilde{S}/\sqrt{\dim C}$, $T = \mathrm{diag}(\theta_a)$, with the canonical $d_a > 0$; $\lambda_a(b) = \tilde{S}_{ab}/d_a$ is the loop of b around a (Equation (10.2)); ω_0 is the killing loop of Lemma 10.6 (not a 3-cocycle). Here c is the central charge modulo 8 (Theorem 10.9), never the braiding, and C is charge conjugation. $\varphi = (1+\sqrt{5})/2$. For $\mathrm{SU}(2)_k$ the labels are spins $j = 0, \frac{1}{2}, \ldots, \frac{k}{2}$ with $d_j = [2j+1]_q$ and $\theta_j = e^{2\pi i\, j(j+1)/(k+2)}$ (Theorem 9.9, after the flip). The toric code's anyons are $1, e, m, \varepsilon$, as in Sheet 7.

Exercise 1 ($(ST)^3$ for Fibonacci). Take Fibonacci's S (Example 10.11), $\theta = \theta_\tau = e^{4\pi i/5}$ and $T = \mathrm{diag}(1, \theta)$.

(i) Compute $\mathrm{tr}(ST)$ and $\det(ST)$, and show that $(\mathrm{tr}\, ST)^2 = \det(ST)$.

(ii) Deduce from Cayley–Hamilton that $(ST)^3 = \dfrac{\theta(1-\theta)}{\sqrt{2+\varphi}}\, 1$, and show that this number is $e^{7\pi i/10}$.

(iii) Compare with $(ST)^3 = e^{2\pi i c/8} S^2$ and the central charge of Example 10.13.

Solution. (i) With $s = 1/\sqrt{2+\varphi}$, $ST = s\begin{pmatrix} 1 & \varphi\theta \\ \varphi & -\theta \end{pmatrix}$, so $t = \mathrm{tr}\, ST = s(1-\theta)$ and $D = \det(ST) = \det S \det T = -\theta$ (as $\det S = -s^2(1+\varphi^2) = -1$). Then $t^2 = D$ means $(1-\theta)^2 = -\theta(2+\varphi)$, that is $1 + \theta^2 = -\varphi\theta$, or $\theta + \theta^{-1} = -\varphi$; and indeed $\theta + \theta^{-1} = 2\cos(4\pi/5) = -\varphi$.

(ii) By Cayley–Hamilton $M^2 = tM - D$ for $M = ST$, so $M^3 = tM^2 - DM = (t^2 - D)M - tD = -tD$, a scalar: $(ST)^3 = s(1-\theta)\theta\, 1$. Now $1 - \theta = -2i\sin(2\pi/5)e^{2\pi i/5}$, so $\theta(1-\theta) = -2i\sin(2\pi/5)e^{6\pi i/5} = 2\sin(2\pi/5)e^{6\pi i/5 - i\pi/2} = 2\sin(2\pi/5)e^{7\pi i/10}$, and $4\sin^2(2\pi/5) = 2(1 - \cos(4\pi/5)) = 2 + \varphi$. So $(ST)^3 = e^{7\pi i/10}\, 1$.

(iii) $S^2 = 1$ (Example 10.12), so the relation of Theorem 10.10 reads $(ST)^3 = e^{2\pi i c/8}$, and $2\pi c/8 = 7\pi/10$ for $c = \frac{14}{5}$: the central charge of the Gauss sum (Example 10.13), now from the modular data alone.

Exercise 2 (The toric code). Take $1, e, m, \varepsilon$ with $\mathbb{Z}_2 \times \mathbb{Z}_2$ fusion ($e \otimes m = \varepsilon$, each its own dual), $d = 1$ and $\theta = (1, 1, 1, -1)$.

(i) Compute S from Proposition 10.3, check $S^2 = 1$, and conclude that the toric code is modular.

(ii) Recover $e \otimes m \cong \varepsilon$ from Verlinde's formula (Theorem 10.8).

(iii) Compute the Gauss sum and c; and the loop $\omega_0 = \frac{1}{4}(1 + e + m + \varepsilon)$ around each anyon.

Solution. (i) Every d is 1 and $a \otimes \bar{b} = a \otimes b$ has one channel, so $\tilde{S}_{ab} = \theta_{a\otimes b}/(\theta_a\theta_b)$. For example $\tilde{S}_{em} = \theta_\varepsilon/(\theta_e\theta_m) = -1$, $\tilde{S}_{e\varepsilon} = \theta_m/(\theta_e\theta_\varepsilon) = -1$, $\tilde{S}_{\varepsilon\varepsilon} = \theta_1/\theta_\varepsilon^2 = 1$, $\tilde{S}_{ee} = \theta_1/\theta_e^2 = 1$, and $\tilde{S}_{1a} = 1$. With $\dim C = 4$, $S = \tilde{S}/2$ is the matrix of Example 10.14. Its rows are orthogonal of length 1, and S is real symmetric, so $S^2 = SS^T = 1 = C$ (all anyons self-dual): S is invertible and the toric code is modular.

(ii) $S_{1x} = \frac{1}{2}$ for all x, so $N_{em}^c = \sum_x \frac{S_{ex}S_{mx}S_{cx}}{S_{1x}} = \frac{1}{4}\sum_x (2S_{ex})(2S_{mx})(2S_{cx})$. The products $2S_{ex} \cdot 2S_{mx}$ over $x = 1, e, m, \varepsilon$ are $(1, -1, -1, 1)$, which is the row $2S_{\varepsilon x}$; so the sum is $\frac{1}{4} \cdot 4 = 1$ for $c = \varepsilon$ and 0 for the other rows, which are orthogonal to it.

(iii) The Gauss sum is $\sum_a d_a^2\theta_a = 1+1+1-1 = 2 = \sqrt{4}\,e^{2\pi ic/8}$ with $c=0$. The loop of b around a is $\lambda_a(b) = \tilde{S}_{ab}$, so ω_0 around a is $\frac{1}{4}\sum_b \tilde{S}_{ab}$: the row sums of $\tilde{S}$, $4,0,0,0$, give 1 around the vacuum and 0 around e, m, ε, as Lemma 10.6 says for a trivial Müger centre.

Exercise 3 (SU(2)$_2$ and SU(2)$_3$). (i) For $k=2$, compute S from Proposition 10.3 with $d=(1,\sqrt{2},1)$ and $\theta=(1,e^{3\pi i/8},-1)$, and compare with the table $S_{jj'} = \sqrt{2/(k+2)}\,\sin\big((2j+1)(2j'+1)\pi/(k+2)\big)$.

(ii) Use Verlinde's formula at $k=2$ to recover $\frac{1}{2}\otimes\frac{1}{2}\cong 0\oplus 1$, and the Gauss sum to find c.

(iii) For $k=3$, write S from the table and use Verlinde's formula to recover $1\otimes 1\cong 0\oplus 1$.

Solution. (i) $\dim C = 1+2+1 = 4$. All labels are self-dual. $\tilde{S}_{0j} = d_j$. $\tilde{S}_{\frac{1}{2}\frac{1}{2}} = (d_0\theta_0 + d_1\theta_1)/\theta_{1/2}^2 = (1-1)/\theta_{1/2}^2 = 0$. $\tilde{S}_{\frac{1}{2}1} = d_{1/2}\theta_{1/2}/(\theta_{1/2}\theta_1) = \sqrt{2}/(-1) = -\sqrt{2}$. $\tilde{S}_{11} = d_0\theta_0/\theta_1^2 = 1$. So $S = \frac{1}{2}\left(\begin{smallmatrix} 1 & \sqrt{2} & 1 \\ \sqrt{2} & 0 & -\sqrt{2} \\ 1 & -\sqrt{2} & 1 \end{smallmatrix}\right)$. The table at $k=2$ has $\sqrt{1/2}\,\sin(mn\pi/4)$ with $m,n\in\{1,2,3\}$: $\sin(\pi/4) = \sin(3\pi/4) = 1/\sqrt{2}$, $\sin(\pi/2)=1$, $\sin(\pi)=0$, $\sin(3\pi/2)=-1$, $\sin(9\pi/4)=1/\sqrt{2}$: the same matrix.

(ii) With $S_{0x} = \frac{1}{2}(1,\sqrt{2},1)$ and $S_{\frac{1}{2}x} = \frac{1}{2}(\sqrt{2},0,-\sqrt{2})$, $S^2_{\frac{1}{2}x}/S_{0x} = (1,0,1)$, so $N^c_{\frac{1}{2}\frac{1}{2}} = \sum_x S^2_{\frac{1}{2}x}S_{cx}/S_{0x} = S_{c0}+S_{c1}$ (the two surviving x): $\frac{1}{2}+\frac{1}{2}=1$ for $c=0$, $\frac{\sqrt{2}}{2}-\frac{\sqrt{2}}{2}=0$ for $c=\frac{1}{2}$, $\frac{1}{2}+\frac{1}{2}=1$ for $c=1$. The Gauss sum is $1+2e^{3\pi i/8}+1\cdot(-1) = 2e^{3\pi i/8} = \sqrt{4}\,e^{2\pi ic/8}$: $c=\frac{3}{2}$, which is $3k/(k+2)$ at $k=2$; this is the $\nu=3$ Ising-type category of Remark 10.16.

(iii) At $k=3$, $S_{jj'} = \sqrt{2/5}\,\sin(mn\pi/5)$ with $m=2j+1$, $n=2j'+1\in\{1,2,3,4\}$; reducing by $\sin(x)=\sin(\pi-x)$ and the period gives, with $a=\sin(\pi/5)$ and $b=\sin(2\pi/5)$,

$$S = \sqrt{\tfrac{2}{5}}\begin{pmatrix} a & b & b & a \\ b & a & -a & -b \\ b & -a & -a & b \\ a & -b & b & -a \end{pmatrix},$$

with $b/a=\varphi$ and $a^2+b^2=\frac{5}{4}$ (Sheet 5's sum $\sum_{m=1}^4 \sin^2(m\pi/5) = \frac{5}{2}$). As S is real and $S_{cx} = S_{0x}\lambda_x(c)$ with $\lambda_x(c) = S_{cx}/S_{0x}$, Verlinde reads $N^c_{11} = \sum_x S^2_{0x}\lambda_x(1)^2\lambda_x(c)$. Over $x=0,\frac{1}{2},1,\frac{3}{2}$: $S^2_{0x} = \frac{2}{5}(a^2,b^2,b^2,a^2)$ and $\lambda_x(1) = (\varphi,-\varphi^{-1},-\varphi^{-1},\varphi)$, so $S^2_{0x}\lambda_x(1)^2 = \frac{2}{5}(b^2,a^2,a^2,b^2)$. With $\lambda_x(0)=(1,1,1,1)$, $\lambda_x(\frac{1}{2}) = (\varphi,\varphi^{-1},-\varphi^{-1},-\varphi)$, $\lambda_x(1)$ as above and $\lambda_x(\frac{3}{2}) = (1,-1,1,-1)$: $N^0_{11} = \frac{4}{5}(a^2+b^2) = 1$; $N^{1/2}_{11} = \frac{2}{5}\big(b^2(\varphi-\varphi)+a^2(\varphi^{-1}-\varphi^{-1})\big) = 0$; $N^1_{11} = \frac{4}{5}(b^2\varphi - a^2\varphi^{-1}) = \frac{4}{5}a^2(\varphi^3-\varphi^{-1}) = \frac{4}{5}a^2(\varphi+2) = 1$, since $b^2=\varphi^2a^2$ and $a^2+b^2=\frac{5}{4}$ give $a^2 = \frac{5}{4(1+\varphi^2)} = \frac{5}{4(2+\varphi)}$; and $N^{3/2}_{11} = \frac{2}{5}\big(b^2(1-1)+a^2(-1+1)\big) = 0$. So $1\otimes 1\cong 0\oplus 1$, Fibonacci's rule; the script checks every entry of both tables.

Exercise 4 (The central charges $\nu/2$). Take the Ising-type data: $1,\sigma,\psi$ with $d=(1,\sqrt{2},1)$, $\theta_\sigma = e^{i\pi\nu/8}$ for odd ν, $\theta_\psi=-1$ (Theorem 7.9, Remark 10.15).

(i) Show that $c=\nu/2$ modulo 8.

(ii) Show that S is the same matrix for every odd ν, and invertible. What, then, tells the eight categories apart?

(iii) Find c for the semion ($d=(1,1)$, $\theta=(1,i)$), and check it against $3k/(k+2)$ at $k=1$.

Solution. (i) $\dim C = 4$ and the Gauss sum is $1\cdot 1+2\theta_\sigma+1\cdot(-1) = 2e^{i\pi\nu/8} = \sqrt{4}\,e^{2\pi ic/8}$ with $2\pi c/8 = \pi\nu/8$, that is $c=\nu/2$ modulo 8.

(ii) $\tilde{S}_{\sigma\sigma} = (d_1\theta_1+d_\psi\theta_\psi)/\theta_\sigma^2 = 0$, $\tilde{S}_{\sigma\psi} = d_\sigma\theta_\sigma/(\theta_\sigma\theta_\psi) = -\sqrt{2}$ and $\tilde{S}_{\psi\psi} = \theta_1/\theta_\psi^2 = 1$: no entry involves ν, and S is the matrix of Exercise 3(i), with $S^2=1$. The eight categories share N, d

and S; only T differs, $\theta_\sigma = e^{i\pi\nu/8}$, and with it $c = \nu/2$. (The indicator $\kappa_\sigma = (-1)^{(\nu^2-1)/8}$ and the F-symbols split the eight only into two fusion categories, four braidings each (Theorem 7.9): $\nu = 1$ is the course's Ising, $\nu = 3$ is $\mathrm{SU}(2)_2$. Only the braiding, through T, separates all eight.) The central charge is defined modulo 8 only: the modular data see $e^{2\pi ic/8}$, and ν and $\nu + 16$ give the same.

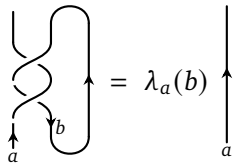
(iii) The semion: $1 + 1 \cdot i = \sqrt{2}\, e^{i\pi/4} = \sqrt{\dim C}\, e^{2\pi ic/8}$ with $c = 1$; and $3k/(k+2) = 1$ at $k = 1$, where $\mathrm{SU}(2)_1$ is the semion (Sheet 9, Exercise 5).

Exercise 5 (Loops and the killing loop). (i) Draw the loop of b around a strand a (Equation (10.2)). For Fibonacci, compute $\lambda_\tau(\tau)$, and check the character relation $\lambda_\tau(\tau)^2 = 1 + \lambda_\tau(\tau)$ of Lemma 10.4.

(ii) Compute the killing loop $\omega_0 = (1 + \varphi\,\tau)/(2 + \varphi)$ around 1 and around τ.

(iii) In sVec $(1, \psi;\ d = (1, 1);\ \theta_\psi = -1;\ \psi \otimes \psi \cong 1)$ compute $\lambda_\psi(\psi)$ and the killing loop around ψ. Which step of the proof of Lemma 10.6 fails, and what does it say about S?

Solution. (i) The picture: the strand a passes up through a closed loop b (two crossings), which equals $\tilde{S}_{ab}/d_a$ times the bare strand:



For Fibonacci, $\lambda_\tau(\tau) = \tilde{S}_{\tau\tau}/d_\tau = -1/\varphi$. Since $\tau \otimes \tau \cong 1 \oplus \tau$, Lemma 10.4 says $\lambda_\tau(\tau)^2 = \lambda_\tau(1) + \lambda_\tau(\tau) = 1 + \lambda_\tau(\tau)$: indeed $\varphi^{-2} = 1 - \varphi^{-1}$, from $\varphi^2 = \varphi + 1$ divided by φ^2.

(ii) Around 1, every loop is its dimension: $(1 + \varphi \cdot \varphi)/(2 + \varphi) = 1$. Around τ, the loop τ is $\lambda_\tau(\tau) = -1/\varphi$: $(1 + \varphi \cdot (-1/\varphi))/(2 + \varphi) = 0$. The killing loop kills τ, as Lemma 10.6 says.

(iii) $\tilde{S}_{\psi\psi} = d_1\theta_1/\theta_\psi^2 = 1$, so $\lambda_\psi(\psi) = 1 = d_\psi$: a ψ loop around ψ is the same as an empty loop, and ψ is transparent. The killing loop $\omega_0 = \frac{1}{2}(1 + \psi)$ around ψ is $\frac{1}{2}(1 + 1) = 1$, not 0. In the proof of Lemma 10.6, $\Omega_\psi \neq 0$ forces $\lambda_\psi(b) = d_b$ for all b, which is true, but the last step, "x is transparent, hence $x = 1$", fails: the Müger centre is all of sVec. Then $S = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, with two equal rows, is singular, as Theorem 10.7 predicts.

Exercise 6 ⋆. *The* $\mathrm{SU}(2)_k$ *S-matrix from loops.* Lecture 10 cites $S_{jj'} = \sqrt{2/(k+2)}\,\sin\big((2j+1)(2j'+1)\pi/(k+2)\big)$. Derive it. Write $\vartheta = \pi/(k+2)$ and $y = (2j'+1)\vartheta$.

(i) From Proposition 10.3, show that the loop of spin $\frac{1}{2}$ around j' is $\lambda_{j'}(\frac{1}{2}) = 2\cos y$.

(ii) Using that $\lambda_{j'}$ is a character (Lemma 10.4) and that $[\frac{1}{2}]$ generates the fusion ring (Sheet 5, Exercise 2), show that $\lambda_{j'}(j) = \sin\big((2j+1)y\big)/\sin y$.

(iii) Conclude $\tilde{S}_{jj'} = \sin\big((2j+1)(2j'+1)\vartheta\big)/\sin\vartheta$, and with Sheet 5's global dimension the formula for S. Why must the pivotal flip of Theorem 9.9 be made first?

Solution. (i) The channels of $\frac{1}{2} \otimes j'$ are $c = j' + \frac{1}{2}$ (if $2j' < k$) and $c = j' - \frac{1}{2}$ (if $j' > 0$), and $\theta_c/(\theta_{1/2}\theta_{j'}) = e^{2\pi i\,(c(c+1) - \frac{3}{4} - j'(j'+1))/(k+2)}$. The exponents are $c(c+1) - \frac{3}{4} - j'(j'+1) = j'$ for $c = j' + \frac{1}{2}$ and $-j' - 1$ for $c = j' - \frac{1}{2}$, so the phases are $e^{2ij'\vartheta} = e^{i(y-\vartheta)}$ and $e^{-i(y+\vartheta)}$. With $d_{j'\pm1/2} = \sin(y \pm \vartheta)/\sin\vartheta$ (and the missing channels having $d = 0$: $\sin(y+\vartheta) = \sin\pi = 0$ at $2j' = k$, $\sin(y-\vartheta) = 0$ at $j' = 0$),

$$\tilde{S}_{\frac{1}{2}j'} = \frac{\sin(y+\vartheta)e^{i(y-\vartheta)} + \sin(y-\vartheta)e^{-i(y+\vartheta)}}{\sin\vartheta}.$$

The imaginary part is $\sin(y+\vartheta)\sin(y-\vartheta)-\sin(y-\vartheta)\sin(y+\vartheta)=0$, and the real part is $\sin(y+\vartheta)\cos(y-\vartheta)+\sin(y-\vartheta)\cos(y+\vartheta)=\sin 2y$. So $\tilde{S}_{\frac{1}{2}j'}=\sin 2y/\sin\vartheta=2\cos y\cdot d_{j'}$ and $\lambda_{j'}(\frac{1}{2})=2\cos y$.

(ii) The truncated rule $\frac{1}{2}\otimes j\cong(j-\frac{1}{2})\oplus(j+\frac{1}{2})$ and the character property give $\lambda(j+\frac{1}{2})=\lambda(\frac{1}{2})\lambda(j)-\lambda(j-\frac{1}{2})$ for $j<\frac{k}{2}$, with $\lambda(0)=1$; since $[\frac{1}{2}]$ generates the ring (Sheet 5), these values determine $\lambda_{j'}$. The numbers $u_j=\sin((2j+1)y)/\sin y$ have $u_0=1$, $u_{1/2}=\sin 2y/\sin y=2\cos y$, and $2\cos y\,u_j=u_{j-1/2}+u_{j+1/2}$ (the identity of Sheet 5, Exercise 2(i), with ϑ replaced by y). So $\lambda_{j'}(j)=u_j$.

(iii) $\tilde{S}_{jj'}=d_{j'}\lambda_{j'}(j)=\frac{\sin y}{\sin\vartheta}\cdot\frac{\sin((2j+1)y)}{\sin y}=\frac{\sin((2j+1)(2j'+1)\vartheta)}{\sin\vartheta}$, using $d_{j'}=\sin y/\sin\vartheta$. Sheet 5's starred exercise gives $\dim \mathrm{SU}(2)_k=(k+2)/(2\sin^2\vartheta)$, so $S_{jj'}=\tilde{S}_{jj'}/\sqrt{\dim}=\sqrt{2/(k+2)}\,\sin((2j+1)(2j'+1)\vartheta)$. The flip matters because Proposition 10.3 is proved for the canonical spherical structure, with every $d_c>0$ (Lecture 10); in the Temperley–Lieb structure $d_{1/2}<0$, and the formula is not available there. (Substituting the Temperley–Lieb d and θ into it anyway would multiply $\tilde{S}_{jj'}$ by $(-1)^{2j+2j'}$, since each $d_c\theta_c$ is unchanged and $\theta_a\theta_b$ picks up that sign; the script checks this pattern.)